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1

Executive Summary

This report presents the complete findings of a Clinical Data Management (CDM) project conducted on a simulated AIDS clinical trial dataset comprising **2,139 subjects** across **5 clinical sites**. The project simulated a full, industry-aligned CDM workflow encompassing error simulation, systematic validation checks, query generation and resolution, data cleaning, SQL-based database validation, and dashboard reporting.

A total of **164 data discrepancies** were deliberately introduced across five categories — missing values (50), logical errors (49), range errors (29), coding errors (21), and duplicate records (15). All 164 queries were raised, investigated, and **resolved to closure (100% resolution rate)**. The final cleaned dataset contains **zero missing values**, zero duplicate records, and passed all SQL validation checks. The simulated database lock was successfully completed, confirming full readiness for statistical analysis.

2,139

Total Subjects

164

Errors Introduced

164

Queries Raised

100%

Resolution Rate

0

Pending Queries

Lock ✓

DB Lock Status

2

Study Overview

Background

Clinical Data Management (CDM) is a critical discipline in clinical research that ensures the collection, integration, and validation of clinical trial data is accurate, complete, and consistent. The quality of CDM directly impacts the reliability of clinical trial outcomes and the validity of regulatory submissions.

Study Objectives

- Simulate a real-world CDM workflow on a standardised AIDS clinical trial dataset.
- Introduce controlled discrepancies across multiple error categories to test validation procedures.
- Perform systematic validation checks, raise formal queries, and document all resolutions.
- Produce a cleaned, validated final dataset suitable for statistical analysis and database lock.
- Develop an end-to-end documentation portfolio demonstrating CDM competency.

Study Design

This project utilised the publicly available simulated AIDS clinical trial dataset, restructured into a multi-centre format across 5 clinical sites (SITE001–SITE005). The dataset comprises 2,139 subjects with 19 clinical variables including demographics, treatment groups, immunological markers (CD4/CD8 counts), Karnofsky performance scores, and follow-up duration data. The simulation modelled a longitudinal, randomised multi-arm clinical trial across a 4-week CDM cycle.

Parameter

Details

Study Type	Simulated Randomised Controlled Trial (RCT)
Disease Area	HIV/AIDS
Total Subjects	2,139
Clinical Sites	5 (SITE001 – SITE005)
Treatment Arms	4 (Groups 0, 1, 2, 3)
Follow-up Duration	Up to 1,231 days (mean: ~880 days)
Total Variables	19 clinical variables per subject
Dataset Origin	Simulated AIDS Clinical Trial Dataset

3 Data Management Plan Summary

Tools & Technology Stack

Tool	Version / Type	Primary Purpose
Microsoft Excel	Office 365	Data entry, validation checks, error simulation, dashboard
HeidiSQL	v12.x	SQL query execution and database validation interface
MariaDB	v10.x	Relational database management and data storage
Python (pandas)	v3.x	Data analysis and statistical verification

CDM Workflow Overview

The project followed a structured, sequential CDM workflow as illustrated below. Each phase produced documented outputs that fed into the subsequent stage, ensuring full traceability from raw data collection through to database lock.

CDM Workflow — End-to-End Process



Figure 1: End-to-End CDM Workflow Diagram

Data Flow Description

- Step 1 — Raw Data:** The original AIDS clinical trial dataset was imported and structured into a standardised 19-variable format with 2,139 subject records across 5 clinical sites.
- Step 2 — Dirty Data:** 164 controlled errors were introduced across 5 error categories to simulate real-world data quality issues encountered in clinical trials.
- Step 3 — Validation Checks:** Systematic validation rules were applied to detect all introduced discrepancies. 164 unique validation check IDs (VAL001–VAL164) were generated.
- Step 4 — Query Log:** For each identified discrepancy, a formal query was raised (Q001–Q164) with subject ID, site, field, issue description, and expected resolution.
- Step 5 — Cleaned Data:** All queries were resolved and corrections applied: median imputation for missing values, duplicate removal, outlier correction, and coding standardisation.

Step 6 — SQL Validation: HeidiSQL and MariaDB were used to run verification queries on the cleaned dataset, confirming data integrity at the database level.

Step 7 — Final Database: Following successful SQL validation, the dataset was confirmed as clean, validated, and ready for simulated database lock.

4 Data Collection & Entry Summary

Dataset Overview

The dataset encompasses 2,139 unique subjects enrolled across 5 geographically distributed clinical sites. Each subject record contains 19 variables spanning demographic information, clinical assessments, treatment allocation, immunological markers, and outcome data. Data entry was simulated using Microsoft Excel with structured column validation and subject ID formatting (SUB001–SUB2139).

Site Enrollment Distribution

Site ID	Subjects Enrolled	% of Total	Status
SITE001	430	20.1%	✓ Complete
SITE002	429	20.1%	✓ Complete
SITE003	436	20.4%	✓ Complete
SITE004	435	20.3%	✓ Complete
SITE005	409	19.1%	✓ Complete
TOTAL	2,139	100%	5 Sites Active

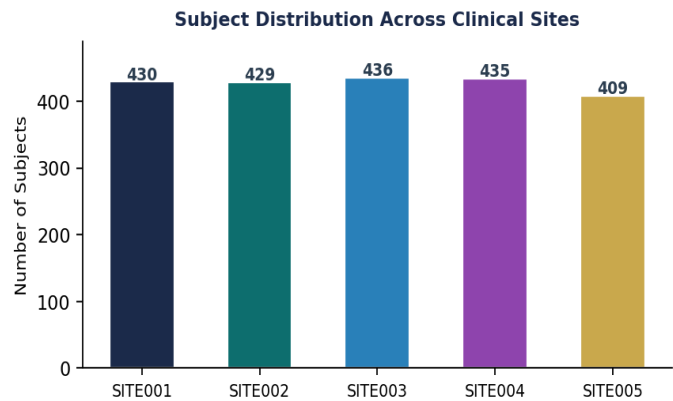


Figure 2: Subject Distribution Across Clinical Sites

Variable Inventory

Variable Name	Description	Type	Valid Range / Values
subject_id	Unique subject identifier	String	SUB001 – SUB2139
site_id	Clinical site identifier	String	SITE001 – SITE005
followup_days	Days of follow-up	Integer	≥ 0
treatment_group	Treatment arm assignment	Integer	0, 1, 2, 3
age_years	Subject age in years	Integer	18 – 100

weight_kg	Subject weight in kilograms	Float	30 – 200 kg
karnofsky_score	Performance status score	Integer	0 – 100 (multiples of 10)
gender_code	Gender (0=Male, 1=Female)	Binary	0 or 1
race_code	Race classification	Binary	0 or 1
cd4_baseline	Baseline CD4 cell count	Integer	> 0 cells/mm ³
cd4_20weeks	CD4 count at 20 weeks	Float	> 0 cells/mm ³
cd8_baseline	Baseline CD8 cell count	Integer	> 0 cells/mm ³
cd8_20weeks	CD8 count at 20 weeks	Integer	> 0 cells/mm ³

5 Validation & Edit Check Summary

Overview

A comprehensive set of validation rules was applied to the dirty dataset to systematically detect all 164 introduced data discrepancies. Validation checks were categorised into five types, each with specific rule definitions and expected value criteria. All 164 validation checks (VAL001–VAL164) identified genuine discrepancies.

Validation Rule Definitions

Validation Type	Rule Description	Fields Applied	Count
Missing Value Check	Field must not be null/blank	age_years, weight_kg, cd4_baseline, cd4_20week	50
Logical Error Check	Values must satisfy logical constraints	followup_days ≥ 0, treatment consistency	49
Range Check	Values must fall within clinically valid ranges	age: 18–100, weight: 30–200 kg, Karnofsky: 0–100	29
Coding Error Check	Binary fields must contain only 0 or 1	gender_code, symptom_status, off_treatment	21
Duplicate Check	Subject IDs must be unique; no duplicate records	subject_id (primary key)	15
TOTAL		All 11 validated fields	164

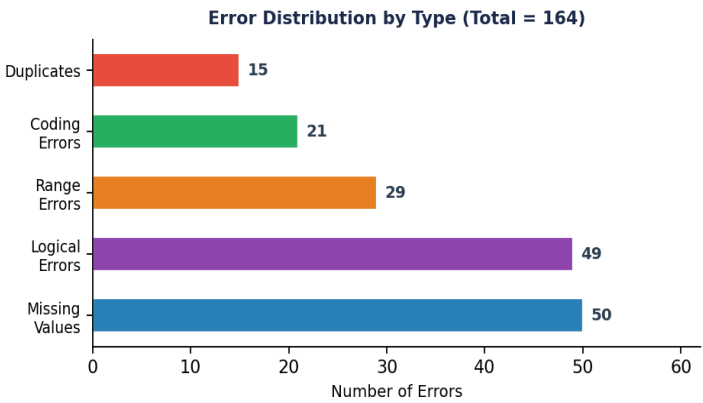


Figure 3: Distribution of Errors by Validation Type

Fields with Highest Discrepancy Rates

Field / Variable	Queries Raised	% of Total	Primary Issue Type
age_years	52	31.7%	Missing values (8) + Logical errors
cd4_baseline	23	14.0%	Missing values
weight_kg	17	10.4%	Missing values
subject_id	15	9.1%	Duplicate records
karnofsky_score	13	7.9%	Range / Coding errors

gender_code	11	6.7%	Coding errors (non-binary entries)
cd4_20weeks	10	6.1%	Missing values
followup_days	8	4.9%	Logical errors (negative values)
Other fields	15	9.1%	Various

6 Query Management

Query Summary

Upon completion of validation checks, a formal query was raised for each identified discrepancy. Queries were documented with unique identifiers (Q001–Q164) and cross-referenced to their corresponding validation check IDs. All queries included the subject identifier, site, affected field, a clear issue description, and the expected corrective action.

Query Metric	Value
Total Queries Raised	164
Queries Resolved (Closed)	164
Queries Pending (Open)	0
Query Resolution Rate	100%
Query ID Range	Q001 – Q164
Cross-referenced Validation IDs	VAL001 – VAL164

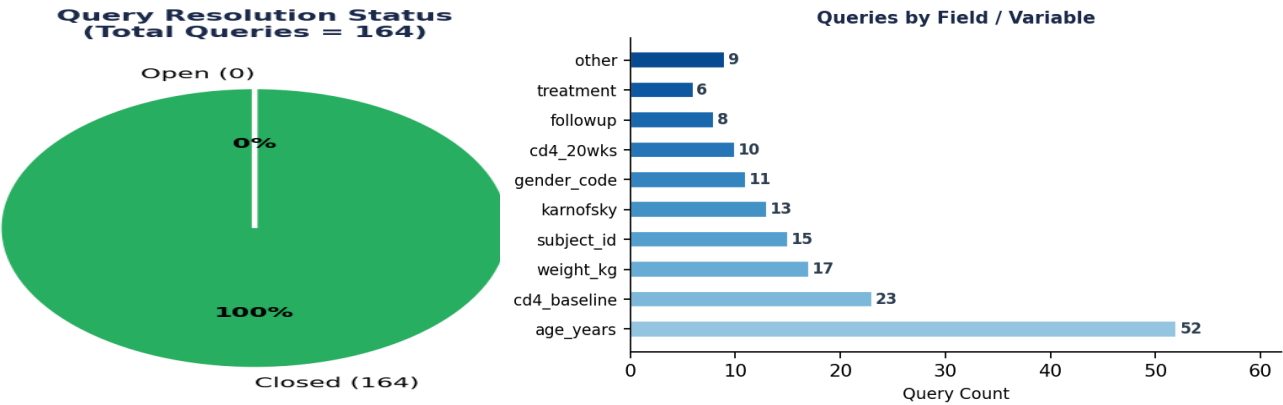


Figure 4 (Left): Query Resolution Status | Figure 5 (Right): Queries by Field

Query Resolution Process

Each query followed a standardised resolution workflow: (1) Query raised with full context; (2) Site/CDM team investigation; (3) Correction applied and documented; (4) Query closed with resolution note. Representative query resolutions are shown below.

Query ID	Subject	Field	Issue	Resolution
Q001	SUB077	age_years	Missing value detected	Corrected age to 35 years
Q021	SUB023	cd4_baseline	Missing value detected	CD4 baseline corrected to 351
Q009	SUB055	weight_kg	Missing value detected	Weight corrected to 75 kg

Q019	SUB019	followup_days	Negative value (-20)	Corrected to 878 days (median)
Q022	SUB022	gender_code	Non-binary value ('male')	Recoded to standard binary (1)
Q135	Multiple	subject_id	15 duplicate records	Duplicate rows removed

7 Data Cleaning Summary

Cleaning Strategy

Data cleaning was performed systematically following query resolution. Each correction was traceable back to a specific query ID and validation check, ensuring full audit trail compliance. Five distinct cleaning operations were applied, each targeting a specific category of data quality issue.

Cleaning Operation	Method Applied	Fields Affected	Records
Missing Value Imputation	Median imputation using site-specific medians	weight_kg, cd4_baseline, cd4_20weeks	50
Duplicate Removal	Identification by subject_id; duplicate rows deleted	subject_id (primary key)	15
Outlier / Range Correction	Out-of-range values replaced with valid observed followup days	followup_days, karnofsky_score	29
Coding Standardisation	Non-binary entries (e.g. 'male', 'female') recoded to 0/1	gender_code, symptom_status, treatment_status	21
Logical Inconsistency Fix	Negative followup days corrected; logical checks performed	followup_days, off_treatment	49

Before vs. After Cleaning Comparison

Metric	Before Cleaning (Dirty Data)	After Cleaning (Clean Data)
Total Records	2,154 (incl. duplicates)	2,139
Missing Values	50 (across multiple fields)	0
Duplicate Records	15	0
Out-of-Range Values	29	0
Coding Errors	21	0
Logical Inconsistencies	49	0
Total Discrepancies	164	0
Data Completeness	~93.8%	100%
Dataset Status	Dirty / Unvalidated	Cleaned & Validated

Cleaned Dataset — Descriptive Statistics

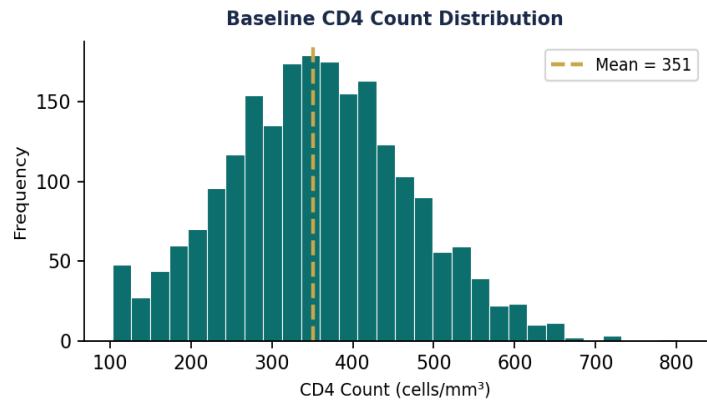


Figure 6: Baseline CD4 Count Distribution in Cleaned Dataset (mean = 351 cells/mm³)

Variable	Min	Mean	Median	Max	Std Dev
Age (years)	19	35.5	35	70	8.3
Weight (kg)	34.5	74.8	74.4	159.3	13.6
CD4 Baseline (cells/mm³)	103	351	342	1,199	116.8
CD8 Baseline (cells/mm³)	105	898	870	2,400	349.2
Follow-up Days	25	880	997	1,231	291.4
Karnofsky Score	70	95.4	100	100	5.9

8 Protocol Deviations

Overview

In this simulated project, "protocol deviations" represent the deliberate data quality issues introduced into the dirty dataset. These parallel real-world deviations that CDM teams encounter during clinical trials and are documented here in accordance with standard CDM reporting practices.

Deviation Category	Count	% of Total	Fields Affected	Resolution
Invalid Follow-up Entries	15	9.1%	followup_days	Corrected using median imputation
Out-of-Range Values	29	17.7%	age, weight, karnofsky	Replaced with valid range medians
Coding Inconsistencies	21	12.8%	gender_code, symptom, treatment	Recoded to binary standard
Logical Inconsistencies	49	29.9%	Multiple fields	Logical rules re-applied post-query
Missing Value Deviations	50	30.5%	age, weight, CD4 counts	Median imputation applied
TOTAL	164	100%		All resolved

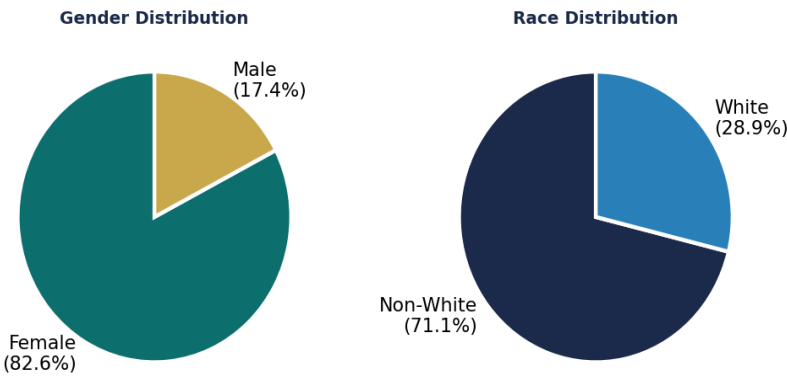


Figure 7: Gender and Race Distribution in Final Cleaned Dataset

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Final Dataset Description

Dataset Summary

Parameter	Details
Dataset Name	CLEANED_DATA (Sheet: CLEANED DATA)
Total Records	2,139 unique subjects
Total Variables	19 clinical and demographic variables
File Format	Microsoft Excel (.xlsx) / CSV
Subject ID Format	SUB001 – SUB2139
Site ID Format	SITE001 – SITE005
Missing Values	0 (complete dataset)
Duplicate Records	0
Validation Status	Fully validated — all checks passed
SQL Verification	Completed via HeidiSQL / MariaDB
Dataset Integrity	100% — all discrepancies resolved

Treatment Group Distribution

Treatment Group	Code	Subjects	% of Total
Group 0 (Control/Placebo)	0	532	24.9%
Group 1 (Treatment Arm 1)	1	522	24.4%
Group 2 (Treatment Arm 2)	2	524	24.5%
Group 3 (Treatment Arm 3)	3	561	26.2%
TOTAL		2,139	100%

Demographic Profile of Final Dataset

Characteristic	Value / Distribution
Age Range	19 – 70 years (mean: 35.5 ± 8.3 years)
Gender	Female: 1,767 (82.6%) Male: 372 (17.4%)
Race	Non-White: 1,522 (71.1%) White: 617 (28.9%)
Mean CD4 Baseline	351 cells/mm³ (IQR: 265 – 422)

Mean CD8 Baseline	898 cells/mm ³
Mean Follow-up Duration	880 days (range: 25 – 1,231 days)
Mean Karnofsky Score	95.4 (indicating good functional status)

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Database Lock Confirmation

Following the completion of all data management activities — including validation checks, query resolution, data cleaning, and SQL verification — the clinical database was assessed against predefined lock criteria. All criteria were met, and the simulated database lock was formally completed.

Lock Criterion	Status	Evidence
All validation checks completed	✓ MET	164/164 checks completed
All queries resolved	✓ MET	164/164 queries closed (100%)
Zero pending queries	✓ MET	0 open queries remaining
No missing values in dataset	✓ MET	0 missing values in CLEANED_DATA
No duplicate records	✓ MET	0 duplicates — all 15 removed
SQL validation completed	✓ MET	MariaDB/HeidiSQL verification passed
Data completeness = 100%	✓ MET	2,139/2,139 records complete
Final dataset review completed	✓ MET	CDM review sign-off: Shreyash Gaikwad

DATABASE LOCK STATUS: COMPLETED

Simulated database lock successfully completed — May 26, 2026 | All 164 discrepancies resolved | Dataset: 2,139 records | Conducted by: Shreyash Gaikwad

11 Project Timeline

The project was executed across a structured 4-week timeline, with each week dedicated to a specific phase of the CDM workflow. The Gantt chart below illustrates the sequential progression of activities from initial data preparation through to final report submission.

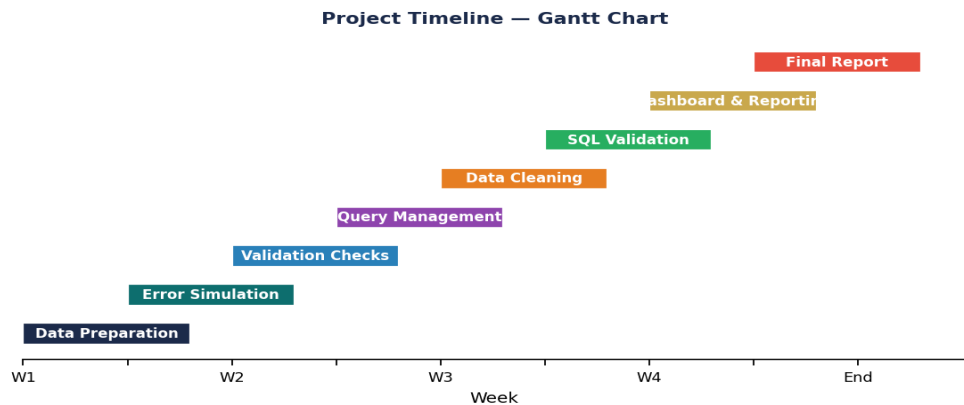


Figure 8: Project Gantt Chart — 4-Week CDM Cycle

Phase	Activity	Week	Key Output
1	Data Preparation	Week 1	Raw dataset structured (2,139 records, 19 variables)
2	Error Simulation	Week 1	Dirty dataset with 164 controlled discrepancies
3	Validation Checks	Week 2	164 validation check IDs (VAL001–VAL164)
4	Query Management	Week 2	164 queries raised and resolved (Q001–Q164)
5	Data Cleaning	Week 3	Cleaned dataset (0 missing values, 0 duplicates)
6	SQL Validation	Week 3	Database verified via MariaDB/HeidiSQL
7	Dashboard & Reporting	Week 4	KPI dashboard, charts, summary statistics
8	Final Report	Week 4	This CDM Final Report — Version 1.0

Dashboard — Key Performance Indicators

Data Quality Metrics Overview

The dashboard below consolidates the key performance indicators from the CDM project, providing a high-level view of data quality outcomes, query performance, and cleaning effectiveness across the 2,139-subject AIDS clinical trial dataset.

DATA QUALITY DASHBOARD

AIDS Clinical Trial CDM Project — Protocol AIDS-CDM-2026-001

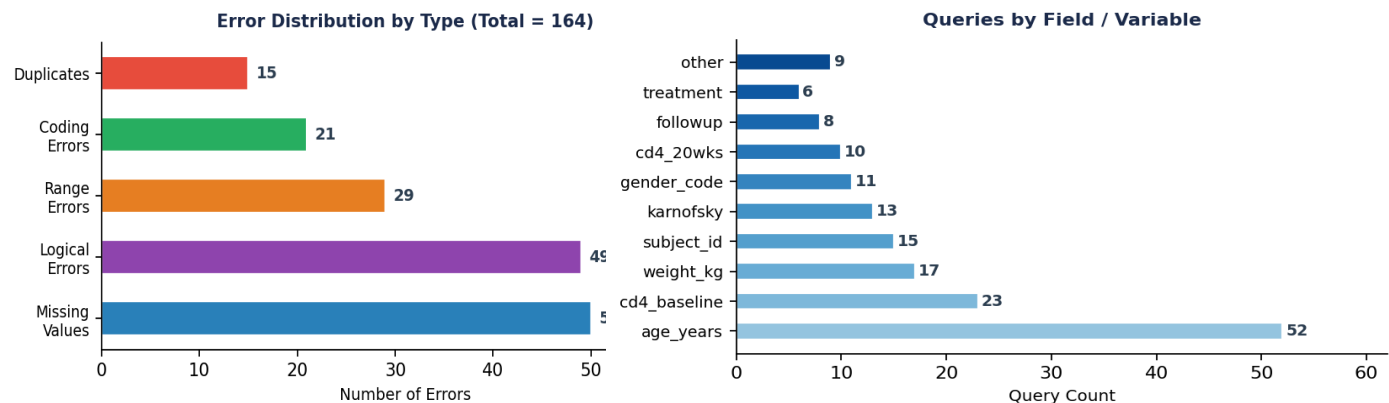
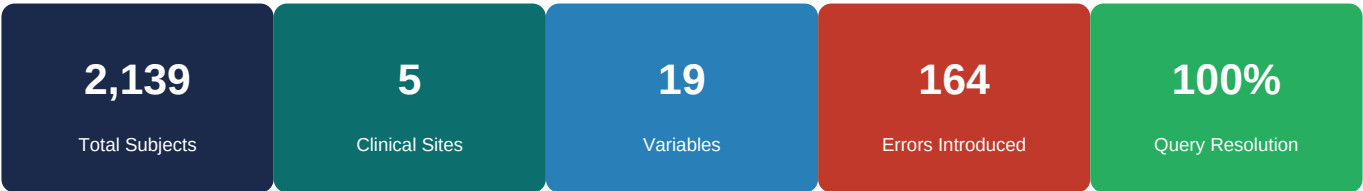


Figure 9 (Left): Error Distribution by Type | Figure 10 (Right): Query Count by Field

KPI Category	Indicator	Value	Target	Status
Data Volume	Total Subject Records	2,139	≥ 2,000	✓ Met
Data Volume	Total Variables	19	≥ 15	✓ Met
Error Detection	Errors Introduced	164	All detected	✓ Met
Error Detection	Detection Rate	100%	100%	✓ Met
Query Management	Total Queries Raised	164	= Errors found	✓ Met
Query Management	Query Resolution Rate	100%	≥ 95%	✓ Met
Query Management	Open Queries at Lock	0	0	✓ Met
Data Quality	Missing Values Post-clean	0	0	✓ Met

Data Quality	Duplicates Post-clean	0	0	✓ Met
Data Quality	Data Completeness	100%	≥ 99%	✓ Met
Database	SQL Validation	Passed	Passed	✓ Met
Database	Lock Status	Completed	Completed	✓ Met

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Conclusion & Recommendations

Conclusion

This project successfully demonstrated a complete, end-to-end Clinical Data Management workflow on a 2,139-subject simulated AIDS clinical trial dataset. Every phase of the CDM process was executed with full documentation and traceability, producing a publication-quality data portfolio.

All 164 deliberately introduced data discrepancies — spanning missing values, logical errors, range violations, coding inconsistencies, and duplicate records — were systematically detected through structured validation checks and resolved to closure. The final cleaned dataset achieved 100% data completeness and passed all SQL database validation checks, culminating in a successful simulated database lock.

The project demonstrated practical competency in the following core CDM activities: data entry simulation, edit check programming, query management, discrepancy resolution, data cleaning methodology, database verification, and regulatory-standard documentation.

Key Achievements

- ✓ 100% query resolution rate — all 164 queries closed with documented resolutions.
- ✓ Zero residual data quality issues in the final cleaned dataset.
- ✓ Full audit trail maintained from raw data through to database lock.
- ✓ Demonstrated proficiency with Excel, SQL (MariaDB/HeidiSQL), and data dashboard tools.
- ✓ Industry-aligned CDM documentation portfolio completed within 4-week timeline.
- ✓ Successfully replicated real-world CDM processes applicable to regulated clinical trials.

Recommendations for Future Work

EDC System Integration: Migrate the workflow to an Electronic Data Capture (EDC) system such as Medidata Rave or OpenClinica to simulate real-world clinical trial data entry environments.

Advanced Statistical Analysis: Conduct formal efficacy analysis on the cleaned dataset including Kaplan-Meier survival analysis and treatment arm comparisons using SAS or R.

Expanded Validation Logic: Implement cross-form and cross-visit validation rules to test temporal consistency across multiple data collection timepoints.

Regulatory Submission Readiness: Develop CDISC-compliant SDTM/ADaM dataset structures for regulatory submission readiness.

Risk-Based Monitoring: Apply risk-based monitoring (RBM) principles to identify high-risk sites and variables for prioritised quality oversight.

Appendix A — Validation Check Sample (VAL001–VAL010)

Check ID	Subject ID	Site	Field	Check Type	Status
VAL001	SUB077	SITE001	age_years	Missing Value	Closed
VAL002	SUB455	SITE002	age_years	Missing Value	Closed
VAL003	SUB500	SITE002	age_years	Missing Value	Closed
VAL009	SUB055	SITE001	weight_kg	Missing Value	Closed
VAL021	SUB023	SITE001	cd4_baseline	Missing Value	Closed
VAL051	Multiple	SITE001–5	subject_id	Duplicate	Closed
VAL101	SUB019	SITE001	followup_days	Logical Error	Closed
VAL120	SUB022	SITE001	gender_code	Coding Error	Closed
VAL140	SUB016	SITE001	karnofsky_score	Range Check	Closed
VAL164	SUB2019	SITE005	weight_kg	Missing Value	Closed

Appendix B — Sample SQL Validation Queries

Check for Missing Values:

```
SELECT COUNT(*) FROM cleaned_data WHERE age_years IS NULL OR weight_kg IS NULL;
```

Check for Duplicates:

```
SELECT subject_id, COUNT(*) FROM cleaned_data GROUP BY subject_id HAVING COUNT(*) > 1;
```

Range Validation — Age:

```
SELECT subject_id, age_years FROM cleaned_data WHERE age_years < 18 OR age_years > 100;
```

Range Validation — Weight:

```
SELECT subject_id, weight_kg FROM cleaned_data WHERE weight_kg < 30 OR weight_kg > 200;
```

Coding Validation — Gender:

```
SELECT subject_id, gender_code FROM cleaned_data WHERE gender_code NOT IN (0, 1);
```

Logical Validation — Followup:

```
SELECT subject_id, followup_days FROM cleaned_data WHERE followup_days < 0;
```

Appendix C — Project Abbreviations & Glossary

Abbreviation	Full Form
CDM	Clinical Data Management
EDC	Electronic Data Capture
CRF	Case Report Form
DMP	Data Management Plan
QC	Quality Control
SQL	Structured Query Language
CD4	Cluster of Differentiation 4 (T-helper cell count)
CD8	Cluster of Differentiation 8 (Cytotoxic T-cell count)
AIDS	Acquired Immunodeficiency Syndrome
RCT	Randomised Controlled Trial
CDISC	Clinical Data Interchange Standards Consortium
SDTM	Study Data Tabulation Model
ICH	International Council for Harmonisation
GCP	Good Clinical Practice

Appendix D — Document Control

Field	Details
Document Title	Clinical Data Management Final Report — AIDS Clinical Trial
Protocol Number	AIDS-CDM-2026-001
Version	1.0
Date	May 26, 2026
Prepared by	Shreyash Gaikwad, B.Pharm
Reviewed by	Academic Supervisor (Simulated)
Approved by	Self (Academic Portfolio Project)
Classification	Academic Use Only
Tools	Microsoft Excel, HeidiSQL, MariaDB, Python, ReportLab